A Simple Workflow for Processing and Visualizing Experimental Data

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Abstract

This short article demonstrates a minimal data workflow for reading raw data, applying basic processing steps, and generating figures for reporting. The pipeline is intentionally lightweight and serves as a template for reproducible analysis and visualization.

1 Introduction

Scientific results often rely on repeatable transformations from raw measurements to processed signals and summary figures [3, 4]. This demo project shows an end-to-end workflow that reads tabular data, applies smoothing and derivative calculations, and produces publication-ready plots [2].

2 Methods

Raw data are read from `data/raw.csv`. The processing script performs the following steps:

- drops rows with missing values,
- detects a time column when available (or uses row index),
- computes a moving-average smooth for numeric columns, and
- computes time derivatives for numeric columns.

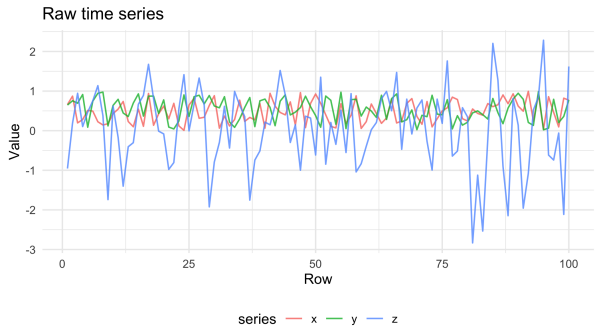
Processed data are written to `data/processed.csv`. The visualization script creates time-series and distribution plots for both raw and processed data, as well as plots for smoothed signals and derivatives when available [1].

3 Results

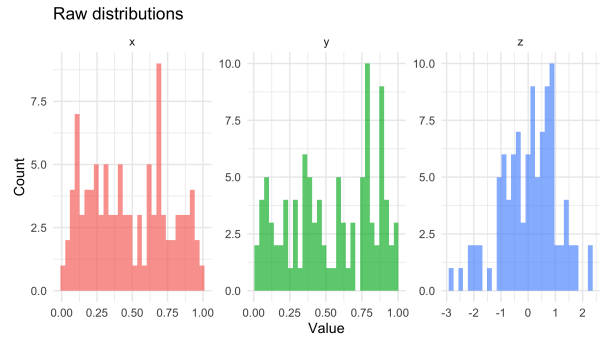
Figure 1 shows raw time-series and distributions. Figure 2 shows advanced visualizations. Figure 3 shows processed time-series, smoothed signals, and time derivatives.

4 Discussion

This demo highlights a reproducible workflow that turns raw tabular data into processed signals and figures suitable for inclusion in a report. The scripts can be extended with domain-specific processing and more refined visualization styles.



(a) Raw time series.



(b) Raw distributions.

Figure 1: Raw data visualizations.

5 Conclusion

The provided scripts and figures form a compact, reproducible analysis pipeline. They can be adapted for small experiments or larger studies with minimal modifications.

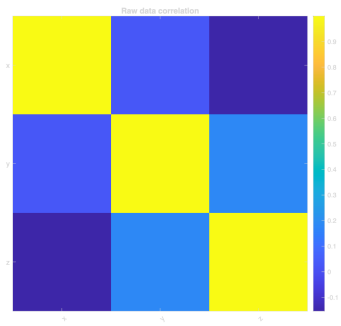
References

[1] Sean McKenna, Alyssa Staub, Brian Stein, and Saleema Amershi. A systematic review of data visualization in science. *IEEE Transactions on Visualization and Computer Graphics*, 27(2):214–223, 2021.

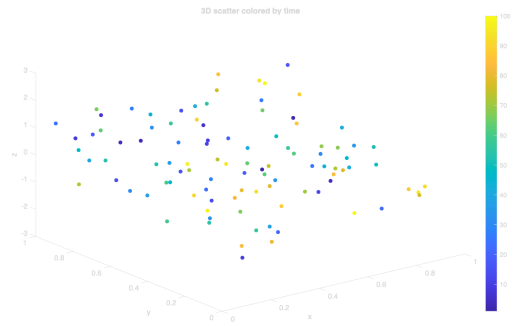
[2] Edward R. Tufte. *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, CT, 2 edition, 2001.

[3] Mark D. Wilkinson, Michel Dumontier, I. J. Aalbersberg, et al. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016.

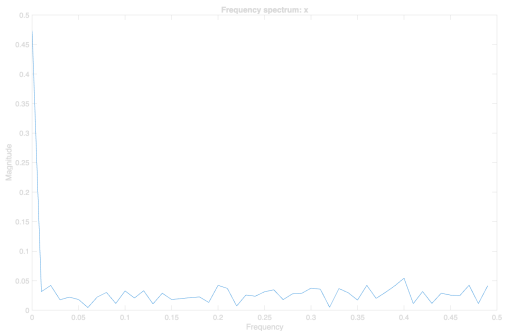
[4] Greg Wilson, Jennifer Bryan, Karen Cranston, et al. Good enough practices in scientific computing. *PLOS Computational Biology*, 15(6):e1005510, 2019.



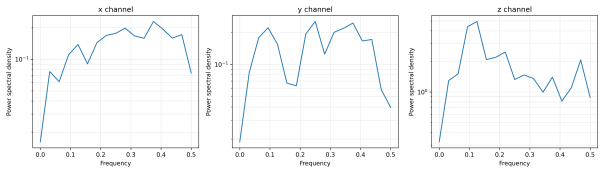
(a) Correlation heatmap.



(b) 3D scatter plot colored by time.

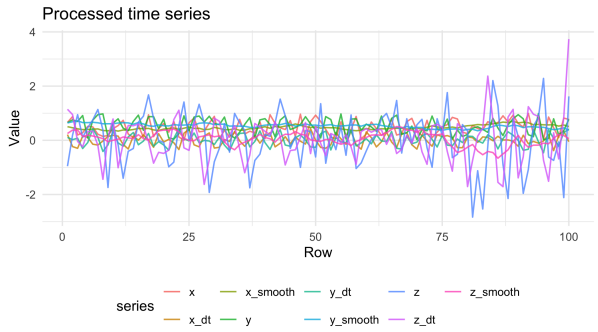


(c) Frequency spectrum.

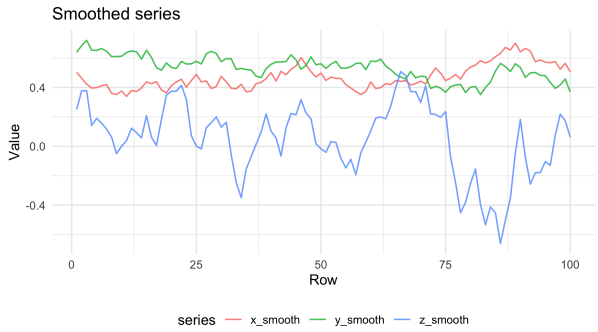


(d) Data spectra.

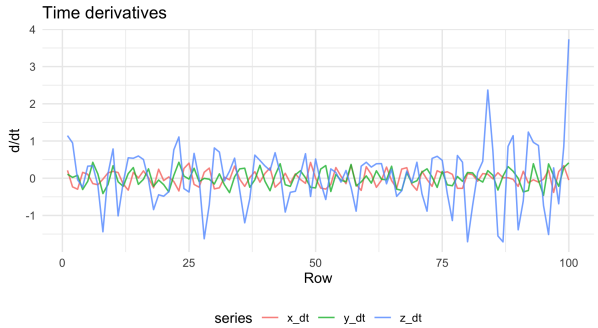
Figure 2: Advanced raw data visualizations.



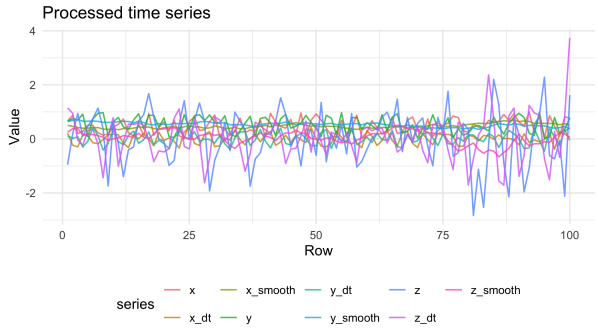
(a) Processed time series.



(b) Smoothed signals.



(c) Time derivatives.



(d) Processed overview (repeat for layout balance).

Figure 3: Processed data visualizations.